

1 (a) Evaluate each of the following.

(i) $(0.008)^{\frac{1}{3}}$

..... [1]

(ii) $\left(\frac{4}{9}\right)^{-\frac{1}{2}}$

..... [1]

(b) Simplify the following expression, giving your answer with positive indices.

$$3x^{-5} \times \frac{2}{3}x^{\frac{1}{2}}$$

..... [2]



2 A manufacturing company purchases a heavy industrial machine for 75000 EUR. The value of the machine depreciates exponentially at a constant rate of 12.5% per year.

(a) Calculate the value of the machine after 4 years. Give your answer to the nearest EUR.

..... [2]

(b) Find the total percentage depreciation in the value of the machine over these 4 years.

..... [2]

3 **(a)** (a) Simplify fully.

$$3\sqrt{48} - \sqrt{12}$$

..... [2]

(b) (b) Rationalise the denominator and simplify fully.

$$\frac{5 + \sqrt{3}}{2 + \sqrt{3}}$$

..... [3]